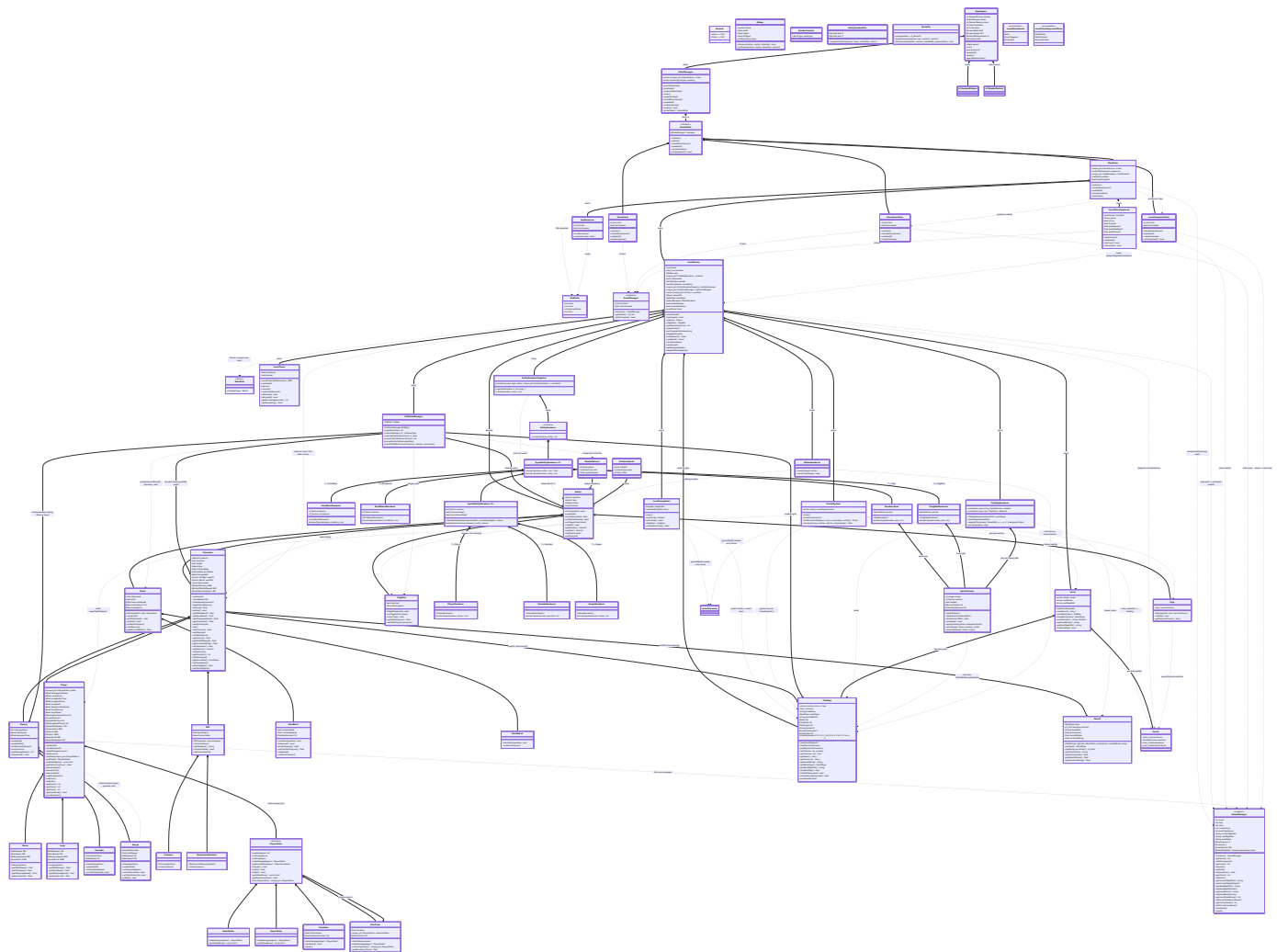


# Full Project Class Diagram

Mermaid diagram covering all model, view, and controller classes (k.2.5). All three restructuring stages are reflected. Stage 1: model interface segregation. Stage 2: the goal castle is painted into the TileMap completion zone from a 21-symbol sheet ( `TileMap::CastleSymbols` , registered by `TileMapRenderer` from atlas rows `y = 696/712` (tower) and `728/744/760` (base)); pipes spawn from 'p' / 'P' grid runs; tile collision treats castle symbols as ground/solid; and every static-image draw goes through the shared `SpritePainter` facade (the `Castle` entity and `CastleRenderer` are gone). Stage 3: the controller god-object split — the live level is `LevelScene` (update reports `Event s`), the flagpole clear play is `LevelClearSequence` , and `PlayState` is slimmed to state transitions, HUD snapshot and debug keys. Stage 3.1: the warp-pipe rules moved out of `LevelScene` into `PortalSystem` (entry detection, one-way inert columns, re-emergence placement), sharing the header-only `controller::geometry helpers ( isGroundSymbol / groundTopAt in LevelGeometry.h )`. Stage 3.2: the goal zone moved out of `LevelScene` into `LevelCompletion` (zone constants, castle paint via `setTile` , flagpole spawn, touch/door queries), leaving `LevelScene` as pure orchestration. Companion doc: `CLASS_ENCAPSULATION_AND_CONNECTIONS.md` (per-class encapsulation + inter-class methods).



## Enums

| Enum            | Values                                                                                                                                                                                                                  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CollisionType   | None, Top, Bottom, Left, Right                                                                                                                                                                                          |
| CollisionLayer  | Player, Enemy, Environment, Trigger                                                                                                                                                                                     |
| AnimState       | Idle, Walk, Run, Jump, Fall, Die                                                                                                                                                                                        |
| PlayerAnimState | SmallIdle, SmallWalk, SmallRun, SmallJump, SmallFall, SmallDie, BigIdle, BigWalk, BigRun, BigJump, BigFall, FireIdle, FireWalk, FireRun, FireJump, FireFall, FireShoot, StarIdle, StarWalk, StarRun, StarJump, StarFall |
| KoopaState      | Walking, ShellIdle, ShellSpinning                                                                                                                                                                                       |
| PortalDirection | Down, Up                                                                                                                                                                                                                |

|                           |                                            |
|---------------------------|--------------------------------------------|
| WorldType                 | Overworld , Underwater , Castle            |
| LevelScene::Event         | None , ClearTriggered , RunEnded           |
| LevelClearSequence::Phase | SlideToPole , WalkToCastle , ReachedCastle |

## Legend

| Arrow | Meaning                            |
|-------|------------------------------------|
| <  -- | Inheritance                        |
| * - - | Composition (owning)               |
| o - - | Aggregation (non-owning reference) |
| ..>   | Dependency / usage                 |